

MongoDB Lab – Day 4 (Aggregation)

Name: Mourima Saha

Roll_no: CSE 326

Schema { roll_no, name, department, semester, city, marks }

Initial Data

Roll No	Name	Department	Semester	City	Marks
1	Amit	CSE	4	Kolkata	78
2	Neha	IT	4	Delhi	65
3	Ravi	CSE	5	Mumbai	82
4	Pooja	ECE	4	Chennai	70
5	Karan	IT	5	Kolkata	55
6	Sneha	CSE	4	Delhi	88

Database Name: college_CSE326

Collection Name: studentlist_CSE326

Lab Questions:

Q1. Display all student records using aggregation.

```
Query: db.studentlist_CSE326.aggregate([  
  { $match: {} }  
])
```

Output: [

```
{  
  _id: ObjectId('69897f3993a870c32b1e2621'),  
  roll_no: 1,  
  name: 'Amit',  
  department: 'CSE',
```

```
semester: 4,  
city: 'Kolkata',  
marks: 78  
},  
{  
  _id: ObjectId('69897f3993a870c32b1e2622'),  
  roll_no: 2,  
  name: 'Neha',  
  department: 'IT',  
  semester: 4,  
  city: 'Delhi',  
  marks: 65  
},  
{  
  _id: ObjectId('69897f3993a870c32b1e2623'),  
  roll_no: 3,  
  name: 'Ravi',  
  department: 'CSE',  
  semester: 5,  
  city: 'Mumbai',  
  marks: 82  
},  
{  
  _id: ObjectId('69897f3993a870c32b1e2624'),  
  roll_no: 4,  
  name: 'Pooja',  
  department: 'ECE',  
  semester: 4,  
  city: 'Chennai',
```

```
marks: 70
},
{
  _id: ObjectId('69897f3993a870c32b1e2625'),
  roll_no: 5,
  name: 'Karan',
  department: 'IT',
  semester: 5,
  city: 'Kolkata',
  marks: 55
},
{
  _id: ObjectId('69897f3993a870c32b1e2626'),
  roll_no: 6,
  name: 'Sneha',
  department: 'CSE',
  semester: 4,
  city: 'Delhi',
  marks: 88
}
]
```

Q2. Display students belonging to CSE department.

```
Query: db.studentlist_CSE326.aggregate([
  { $match: { department: "CSE" } }
])
```

Output: [

```
{
  _id: ObjectId('69897f3993a870c32b1e2621'),
  roll_no: 1,
```

```
name: 'Amit',
department: 'CSE',
semester: 4,
city: 'Kolkata',
marks: 78
},
{
  _id: ObjectId('69897f3993a870c32b1e2623'),
  roll_no: 3,
  name: 'Ravi',
  department: 'CSE',
  semester: 5,
  city: 'Mumbai',
  marks: 82
},
{
  _id: ObjectId('69897f3993a870c32b1e2626'),
  roll_no: 6,
  name: 'Sneha',
  department: 'CSE',
  semester: 4,
  city: 'Delhi',
  marks: 88
}]
```

Q3. Display students of semester 4.

```
Query: db.studentlist_CSE326.aggregate([
  { $match: { semester: 4 } }
])
```

Output: [

```
{
  _id: ObjectId('69897f3993a870c32b1e2621'),
  roll_no: 1,
  name: 'Amit',
  department: 'CSE',
  semester: 4,
  city: 'Kolkata',
  marks: 78
},
{
  _id: ObjectId('69897f3993a870c32b1e2622'),
  roll_no: 2,
  name: 'Neha',
  department: 'IT',
  semester: 4,
  city: 'Delhi',
  marks: 65
},
{
  _id: ObjectId('69897f3993a870c32b1e2624'),
  roll_no: 4,
  name: 'Pooja',
  department: 'ECE',
  semester: 4,
  city: 'Chennai',
  marks: 70
},
{
```

```
_id: ObjectId('69897f3993a870c32b1e2626'),
roll_no: 6,
name: 'Sneha',
department: 'CSE',
semester: 4,
city: 'Delhi',
marks: 88
}]
```

Q4. Display only name and marks of all students.

```
Query: db.studentlist_CSE326.aggregate([
  { $project: { _id: 0, name: 1, marks: 1 } }
])
```

Output: [

```
{ name: 'Amit', marks: 78 },
{ name: 'Neha', marks: 65 },
{ name: 'Ravi', marks: 82 },
{ name: 'Pooja', marks: 70 },
{ name: 'Karan', marks: 55 },
{ name: 'Sneha', marks: 88 }
```

]

Q5. Find the average marks of all students.

```
Query: db.studentlist_CSE326.aggregate([
  { $group: { _id: null, averageMarks: { $avg: "$marks" } } }
])
```

Output: [{ _id: null, averageMarks: 73 }]

Q6. Find total marks scored department-wise.

```
Query: db.studentlist_CSE326.aggregate([
  { $group: { _id: "$department", totalMarks: { $sum: "$marks" } } }
])
```

Output: [

```
{_id: 'ECE', totalMarks: 70 },  
{_id: 'CSE', totalMarks: 248 },  
{_id: 'IT', totalMarks: 120 }
```

]

Q7. Find average marks department-wise.

Query: db.studentlist_CSE326.aggregate([
 { \$group: { _id: "\$department", avgMarks: { \$avg: "\$marks" } } }
])

Output: [

```
{_id: 'ECE', avgMarks: 70 },  
{_id: 'CSE', avgMarks: 82.66666666666667 },  
{_id: 'IT', avgMarks: 60 }
```

]

Q8. Find maximum marks scored in each department.

Query: db.studentlist_CSE326.aggregate([
 { \$group: { _id: "\$department", maxMarks: { \$max: "\$marks" } } }
])

Output: [

```
{_id: 'IT', maxMarks: 65 },  
{_id: 'CSE', maxMarks: 88 },  
{_id: 'ECE', maxMarks: 70 }
```

]

Q9. Count number of students in each department.

Query: db.studentlist_CSE326.aggregate([
 { \$group: { _id: "\$department", totalStudents: { \$sum: 1 } } }
])

Output: [

```
{_id: 'IT', totalStudents: 2 },
```

```
{ _id: 'CSE', totalStudents: 3 },
```

```
{ _id: 'ECE', totalStudents: 1 }]
```

Q10. Display departments having average marks greater than 70.

Query: db.studentlist_CSE326.aggregate([

```
{ $group: { _id: "$department", avgMarks: { $avg: "$marks" } } },
```

```
{ $match: { avgMarks: { $gt: 70 } } }
```

])

Output: [{ _id: 'CSE', avgMarks: 82.66666666666667 }]

Q11. Sort students by marks in descending order.

Query: db.studentlist_CSE326.aggregate([

```
{ $sort: { marks: -1 } }
```

])

Output: [

```
{
```

```
  _id: ObjectId('69897f3993a870c32b1e2626'),
```

```
  roll_no: 6,
```

```
  name: 'Sneha',
```

```
  department: 'CSE',
```

```
  semester: 4,
```

```
  city: 'Delhi',
```

```
  marks: 88
```

```
},
```

```
{
```

```
  _id: ObjectId('69897f3993a870c32b1e2623'),
```

```
  roll_no: 3,
```

```
  name: 'Ravi',
```

```
  department: 'CSE',
```

```
  semester: 5,
```

```
  city: 'Mumbai',
```



```
marks: 82
},
{
  _id: ObjectId('69897f3993a870c32b1e2621'),
  roll_no: 1,
  name: 'Amit',
  department: 'CSE',
  semester: 4,
  city: 'Kolkata',
  marks: 78
},
{
  _id: ObjectId('69897f3993a870c32b1e2624'),
  roll_no: 4,
  name: 'Pooja',
  department: 'ECE',
  semester: 4,
  city: 'Chennai',
  marks: 70
},
{
  _id: ObjectId('69897f3993a870c32b1e2622'),
  roll_no: 2,
  name: 'Neha',
  department: 'IT',
  semester: 4,
  city: 'Delhi',
  marks: 65
},
```

```
{
  _id: ObjectId('69897f3993a870c32b1e2625'),
  roll_no: 5,
  name: 'Karan',
  department: 'IT',
  semester: 5,
  city: 'Kolkata',
  marks: 55
}
```

Q12. Display top 3 students based on marks.

```
Query: db.studentlist_CSE326.aggregate([
  { $sort: { marks: -1 } },
  { $limit: 3 }
])
```

Output: [

```
{
  _id: ObjectId('69897f3993a870c32b1e2626'),
  roll_no: 6,
  name: 'Sneha',
  department: 'CSE',
  semester: 4,
  city: 'Delhi',
  marks: 88
},
{
  _id: ObjectId('69897f3993a870c32b1e2623'),
  roll_no: 3,
  name: 'Ravi',
  department: 'CSE',
```

```

semester: 5,
city: 'Mumbai',
marks: 82
},
{
  _id: ObjectId('69897f3993a870c32b1e2621'),
  roll_no: 1,
  name: 'Amit',
  department: 'CSE',
  semester: 4,
  city: 'Kolkata',
  marks: 78
}]

```

Q13. Display department-wise average marks sorted in descending order.

```

Query: db.studentlist_CSE326.aggregate([
... { $group: { _id: "$department", avgMarks: { $avg: "$marks" } } },
... { $sort: { avgMarks: -1 } }
... ])

```

Output: [

```

{ _id: 'CSE', avgMarks: 82.66666666666667 },
{ _id: 'ECE', avgMarks: 70 },
{ _id: 'IT', avgMarks: 60 }
]

```

Q14. Display students belonging to Delhi or Kolkata.

```

Query: db.studentlist_CSE326.aggregate([{$match:{city:{$in:["Delhi","Kolkata"]}}}
])

```

Output: [

```

{
  _id: ObjectId('69897f3993a870c32b1e2621'),

```

```
roll_no: 1,  
name: 'Amit',  
department: 'CSE',  
semester: 4,  
city: 'Kolkata',  
marks: 78  
,  
{  
  _id: ObjectId('69897f3993a870c32b1e2622'),  
  roll_no: 2,  
  name: 'Neha',  
  department: 'IT',  
  semester: 4,  
  city: 'Delhi',  
  marks: 65  
,  
{  
  _id: ObjectId('69897f3993a870c32b1e2625'),  
  roll_no: 5,  
  name: 'Karan',  
  department: 'IT',  
  semester: 5,  
  city: 'Kolkata',  
  marks: 55  
,  
{  
  _id: ObjectId('69897f3993a870c32b1e2626'),  
  roll_no: 6,  
  name: 'Sneha',
```

department: 'CSE',
semester: 4,
city: 'Delhi',
marks: 88
}]

```
_id: ObjectId('69897f3993a870c32b1e2621')  
roll_no : 1  
name : "Amit"  
department : "CSE"  
semester : 4  
city : "Kolkata"  
marks : 78
```

```
_id: ObjectId('69897f3993a870c32b1e2622')  
roll_no : 2  
name : "Neha"  
department : "IT"  
semester : 4  
city : "Delhi"  
marks : 65
```

```
_id: ObjectId('69897f3993a870c32b1e2623')  
roll_no : 3  
name : "Ravi"  
department : "CSE"  
semester : 5  
city : "Mumbai"  
marks : 82
```

```
_id: ObjectId('69897f3993a870c32b1e2624')  
roll_no : 4  
name : "Pooja"  
department : "ECE"  
semester : 4  
city : "Chennai"  
marks : 70
```

```
_id: ObjectId('69897f3993a870c32b1e2625')  
roll_no : 5  
name : "Karan"  
department : "IT"  
semester : 5  
city : "Kolkata"  
marks : 55
```

```
_id: ObjectId('69897f3993a870c32b1e2626')  
roll_no : 6  
name : "Sneha"  
department : "CSE"  
semester : 4  
city : "Delhi"  
marks : 88
```